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Phong Diep; J. Karl Johnson

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Erratum: ''An accurate H_2–H_2 interaction potential from first principles'' [J. Chem. Phys. 112, 4465 (2000)]

Phong Diep and J. Karl Johnson[a)]
Department of Chemical & Petroleum Engineering, The University of Pittsburgh, Pittsburgh, Pennsylvania 15261

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The values of the second virial coefficient computed using the fit to the *ab initio* H_2–H_2 potential of Diep *et al.* and presented in Fig. 7 [J. Chem. Phys. **112**, 4469 (2000)] are in fortuitous agreement with experimental measurements due to errors in calculation. The main problem was an error in a conversion factor resulting in the significant underestimation of the first-order translational quantum mechanical correction. In addition, the calculations included the ''ideal-gas'' statistical correction, which has been shown to be a valid contribution only at very low temperatures.[1,2] Equation (9) in our previous article[3] should thus be changed to

$$B_2(T)=B_{\mathrm{cl}}(T)+\left(\frac{h^2}{m}\right)B_{\mathrm{I}}(T)+\left(\frac{h^2}{m}\right)^2 B_{\mathrm{II}}(T)+\cdots . \qquad (1)$$

The fit to the *ab initio* potential energy surface (PES) presented in Ref. 3 was slightly biased toward the repulsive energy portion of the potential. The corrected potential fit presented here excludes all points representing the closest H_2–H_2 separation distance of 2.0 Å. A fitting scheme that minimizes the root-mean-square error[4] was employed that minimizes the bias toward extremely high energies of the repulsive walls in the PES. The corrected $G_{224}(\theta_1,\theta_2,\phi_{12})$ term results in a monomer quadrupole moment of 0.650 D Å, which is only slightly higher than the value of 0.649 D Å reported by Poll and Wolniewicz[5] and in better agreement than the value of 0.646 D Å from Ref. 3. The isotropic potential well depth is lowered by 1.82 K, compared to our previous fit. The new parameters $V_{000}(R)$, $V_{022}(R)$, $V_{202}(R)$, and $V_{224}(R)$ are listed in Table I.

Figure 7 in our previous article should be replaced by the corrected results presented in Fig. 1. The correction has shifted all four second virial coefficients in the positive direction. The resulting $B(T)$ computed using our newly fitted potential and that by Wind and Røeggen[6] are slightly higher than that measured experimentally.[7] The calculations also show that the semiclassical method performs significantly better at lower temperatures than previously presented. Figure 2 shows that the semiclassical approximation agrees with both the path integral Monte Carlo method and experimental

TABLE I. Radial components of spherical harmonics interaction model. V_{000} is the isotropic potential. Distances are in angstrom, and energies are in kelvin.

R_{cm}	V_{000}	$V_{022}=V_{202}$	V_{224}
2.00	3514.70	118.87	53.35
2.25	1427.14	51.81	29.60
2.50	519.52	20.77	17.48
2.75	149.75	7.06	10.85
3.00	13.22	1.49	7.03
3.25	−28.26	−0.45	4.71
3.50	−34.43	−0.92	3.25
3.75	−29.57	−0.87	2.30
4.00	−22.71	−0.68	1.67
4.25	−16.69	−0.50	1.23
4.50	−12.08	−0.35	0.93
4.75	−8.74	−0.25	0.71
5.00	−6.38	−0.18	0.55
5.25	−4.72	−0.13	0.43
5.50	−3.54	−9.58E−2	0.34
5.75	−2.69	−7.16E−2	0.27
6.00	−2.07	−5.43E−2	0.22
6.25	−1.61	−4.17E−2	0.18
6.50	−1.27	−3.24E−2	0.15
6.75	−1.01	−2.54E−2	0.12
7.00	−0.80	−2.02E−2	0.10
7.50	−0.53	−1.30E−2	7.19E−2
8.00	−0.36	−8.69E−3	5.21E−2
8.50	−0.25	−5.95E−3	3.85E−2
9.00	−0.17	−4.17E−3	2.89E−2
9.50	−0.13	−2.98E−3	2.21E−2
10.00	−0.09	−2.24E−3	1.75E−2

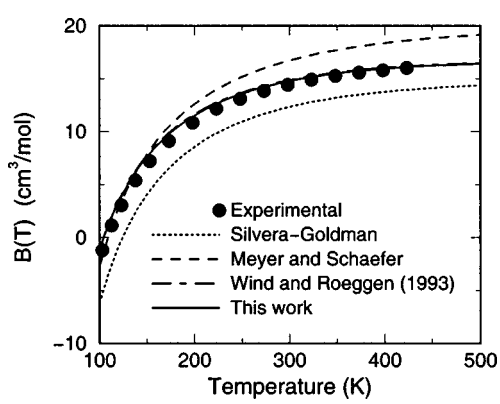

FIG. 1. Second virial coefficients calculated from isotropic $(H_2)_2$ potentials. Quantum mechanical effects were included using a semiclassical approximation.

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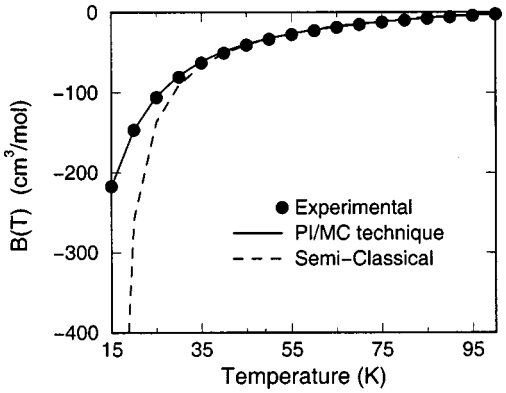

FIG. 2. Second virial coefficient from experiment and calculated using the *ab initio* potential of this work. Calculated values from the semiclassical (dashed) and full path integral (solid) expressions are shown.

data down to about 50 K. It should be stressed that the errors in Ref. 3 are not related to the accuracy of the *ab initio* PES.

[a]Author to whom all correspondence should be addressed. Electronic mail: karlj@pitt.edu

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